EX No:03

Date:

Build a Multiple Linear Regression Model

3.This question involves the use of multiple linear regression on the Auto data set.

LOAD LIBRARY:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from ISLP import load_data
from ISLP.models import (ModelSpec as MS, summarize, poly)
```

LOAD DATA:

```
Auto=load_data("Auto")
```

(a) Produce a scatterplot matrix which includes all of the variables in the data set.

CODE:

```
sns.pairplot(Auto);
```

OUTPUT:



(b) Compute the matrix of correlations between the variables using the DataFrame.corr() method.

CODE:

```
Auto.corr(numeric_only=True)
```

OUTPUT:

	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin
mpg	1.000000	-0.777618	-0.805127	-0.778427	-0.832244	0.423329	0.580541	0.565209
cylinders	-0.777618	1.000000	0.950823	0.842983	0.897527	-0.504683	-0.345647	-0.568932
displacement	-0.805127	0.950823	1.000000	0.897257	0.932994	-0.543800	-0.369855	-0.614535
horsepower	-0.778427	0.842983	0.897257	1.000000	0.864538	-0.689196	-0.416361	-0.455171
weight	-0.832244	0.897527	0.932994	0.864538	1.000000	-0.416839	-0.309120	-0.585005
acceleration	0.423329	-0.504683	-0.543800	-0.689196	-0.416839	1.000000	0.290316	0.212746
year	0.580541	-0.345647	-0.369855	-0.416361	-0.309120	0.290316	1.000000	0.181528
origin	0.565209	-0.568932	-0.614535	-0.455171	-0.585005	0.212746	0.181528	1.000000

(c) Use the sm.OLS() function to perform a multiple linear regression with mpg as the response and all other variables except name as the predictors. Use the summarize () function to print the results.

CODE:

```
X = sm.add_constant(Auto.drop(columns='mpg'))
y = Auto['mpg']
model= sm.OLS(y,X).fit()
print(model.summary())
```

OUTPUT:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    mpg   R-squared:                       0.821
Model:                            OLS   Adj. R-squared:                  0.818
Method:                 Least Squares   F-statistic:                     252.4
Date:                Tue, 13 Oct 2026   Prob (F-statistic):          2.04e-139
Time:                        19:09:20   Log-Likelihood:                -1023.5
No. Observations:                 392   AIC:                             2063.
Df Residuals:                     384   BIC:                             2095.
Df Model:                           7
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const          -17.2184      4.644     -3.707      0.000     -26.350      -8.087
cylinders       -0.4934      0.323     -1.526      0.128      -1.129       0.142
displacement     0.0199      0.008      2.647      0.008       0.005       0.035
horsepower      -0.0170      0.014     -1.230      0.220      -0.044       0.010
weight          -0.0065      0.001     -9.929      0.000      -0.008      -0.005
acceleration     0.0806      0.099      0.815      0.415      -0.114       0.275
year             0.7508      0.051     14.729      0.000       0.651       0.851
origin           1.4261      0.278      5.127      0.000       0.879       1.973
==============================================================================
Omnibus:                       31.906   Durbin-Watson:                   1.309
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               53.100
...
```

i. Is there a relationship between the predictors and the response? Use the anova_lm() function from statsmodels to answer this question.

CODE:

```
from statsmodels.stats.anova import anova_lm
design = MS()
X = design.fit_transform(Auto)
y = Auto['mpg']
results1 = sm.OLS(y, X).fit()
anova_lm(results1, model)
```

OUTPUT:

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	391.0	23818.993469	0.0	NaN	NaN	NaN
1	384.0	4252.212530	7.0	19566.780939	252.428045	2.037106e-139

ii. Which predictors appear to have a statistically significant relationship to the response?

ANS:

Displacement, weight, year, and origin all appear to have a statistically significant relationship to the response "mpg". This is indicated by their small p-values.

iii. What does the coefficient for the year variable suggest?

ANS:

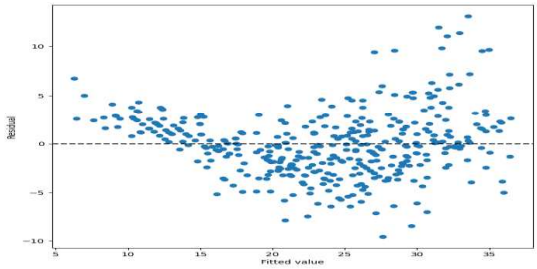
The coefficient for the "year" variable suggests that each year, on average cars "mpg" increases by 0.75

(d) Produce some of diagnostic plots of the linear regression fit as described in the lab. Comment on any problems you see with the fit. Do the residual plots suggest any unusually large outliers? Does the leverage plot identify any observations with unusually high leverage?

CODE:

```
fig, ax = plt.subplots(figsize=(8,8))
plt.scatter(x=results.fittedvalues, y=results.resid)
plt.xlabel('Fitted value')
plt.ylabel('Residual')
plt.axhline(0, c='k', ls='--')
plt.show()
```

OUTPUT:



(e) Fit some models with interactions as described in the lab. Do any interactions appear to be statistically significant?

CODE:

```
allvars = list(Auto.columns.drop('mpg'))
y = Auto['mpg']
final = allvars + [('weight', 'acceleration')]
X = MS(final).fit_transform(Auto)
results2 = sm.OLS(y, X).fit()
summarize(results2)
```

OUTPUT:

	coef	std err	t	P>|t|
intercept	-43.6409	5.811000	-7.511	0.000
cylinders	-0.2141	0.308000	-0.696	0.487
displacement	0.0031	0.007000	0.419	0.676
horsepower	-0.0414	0.013000	-3.071	0.002
weight	0.0040	0.002000	2.462	0.014
acceleration	1.6292	0.242000	6.726	0.000
year	0.7821	0.048000	16.184	0.000
origin	1.0329	0.269000	3.846	0.000
weight:acceleration	-0.0006	0.000084	-6.928	0.000

CODE:

```
sm.stats.anova_lm(results, results2)
```

OUTPUT:

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	384.0	4252.212530	0.0	NaN	NaN	NaN
1	383.0	3778.654965	1.0	473.557565	47.999235	1.813073e-11

INTERPRETATION:

- (a), (b) mpg is strongly negatively correlated with weight (-0.83), displacement (-0.81), horsepower (-0.78) and cylinders (-0.78), and positively with year (0.58), origin (0.57) and acceleration (0.42). The predictors are also highly collinear with each other: displacement-cylinders 0.95, displacement-weight 0.93, displacement-horsepower 0.90.
- (c)(i) Yes, there is a relationship between the predictors and mpg. The full model has R-squared 0.821 (adjusted 0.818) and F = 252.4 with p of about 2e-139; the ANOVA against the intercept-only model cuts the residual sum of squares from 23,819 to 4,252. The model explains about 82% of the variation in mpg.
- (c)(ii) At the 5% level, weight (-0.0065, $p < 0.001$), year (0.7508, $p < 0.001$), origin (1.4261, $p < 0.001$) and displacement (0.0199, $p = 0.008$) are statistically significant. Cylinders ($p = 0.128$), horsepower ($p = 0.220$) and acceleration ($p = 0.415$) are not, even though cylinders and horsepower correlate strongly with mpg on their own, because collinearity with weight and displacement masks their individual effect.
- (c)(iii) The year coefficient of about 0.75 means that, holding the other predictors fixed, mpg increases by roughly 0.75 for every later model year (about 7.5 mpg per decade), reflecting steady improvements in fuel efficiency.
- (d) The residuals-versus-fitted plot is U-shaped, so the true relationship is not fully linear, and the spread of residuals widens at higher fitted values (non-constant variance). Several observations have unusually large positive residuals (about +10 to +13) at fitted values near 30 to 35. A leverage plot was not produced in this record, so high-leverage points are not assessed.
- (e) The weight x acceleration interaction is statistically significant (coefficient -0.0006, $p < 0.001$). Adding it lowers the residual sum of squares from 4,252 to 3,779 (F = 48.0, p of about 1.8e-11), and horsepower ($p = 0.002$), weight ($p = 0.014$) and acceleration ($p < 0.001$) become significant. The effect of weight on mpg therefore depends on acceleration, and the interaction model is a better fit.